

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A Data-Driven Overview • 1750 – 2024

Data: Global Carbon Budget 2025 | NASA | Our World in Data

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

February 2026

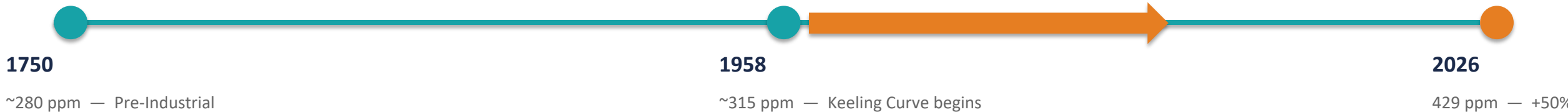
Mauna Loa Observatory (NOAA)

Pre-Industrial Baseline

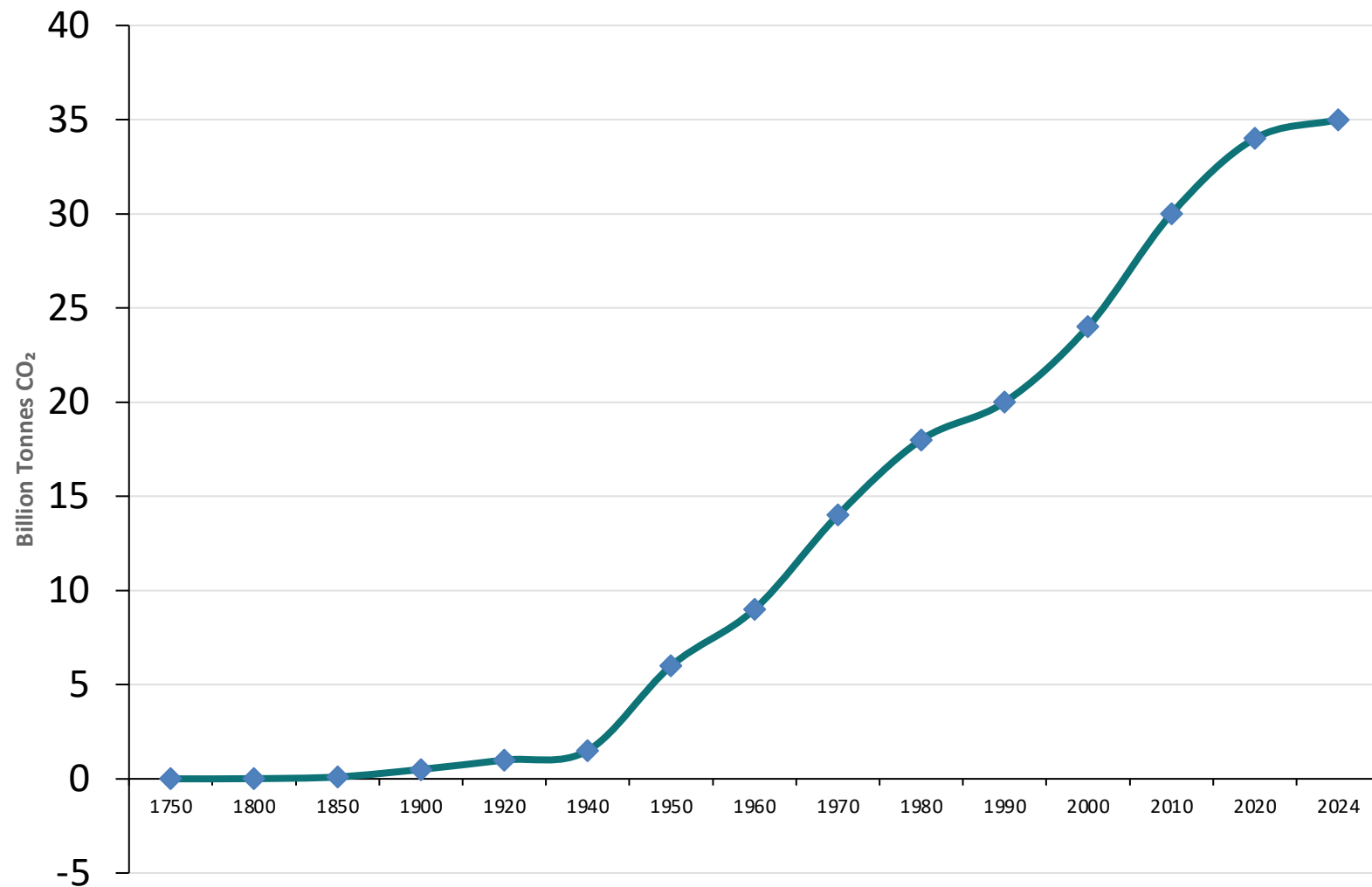
~280 ppm — the stable level for millennia before the Industrial Revolution

The 429 ppm reading is at least 1.5× greater than the natural CO₂ increase observed at the end of the last ice age 20,000 years ago — a change that took thousands of years then, but has occurred in roughly two centuries now.

Acceleration since 1958



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



1950: ~6 Bt

Post-war industrial expansion

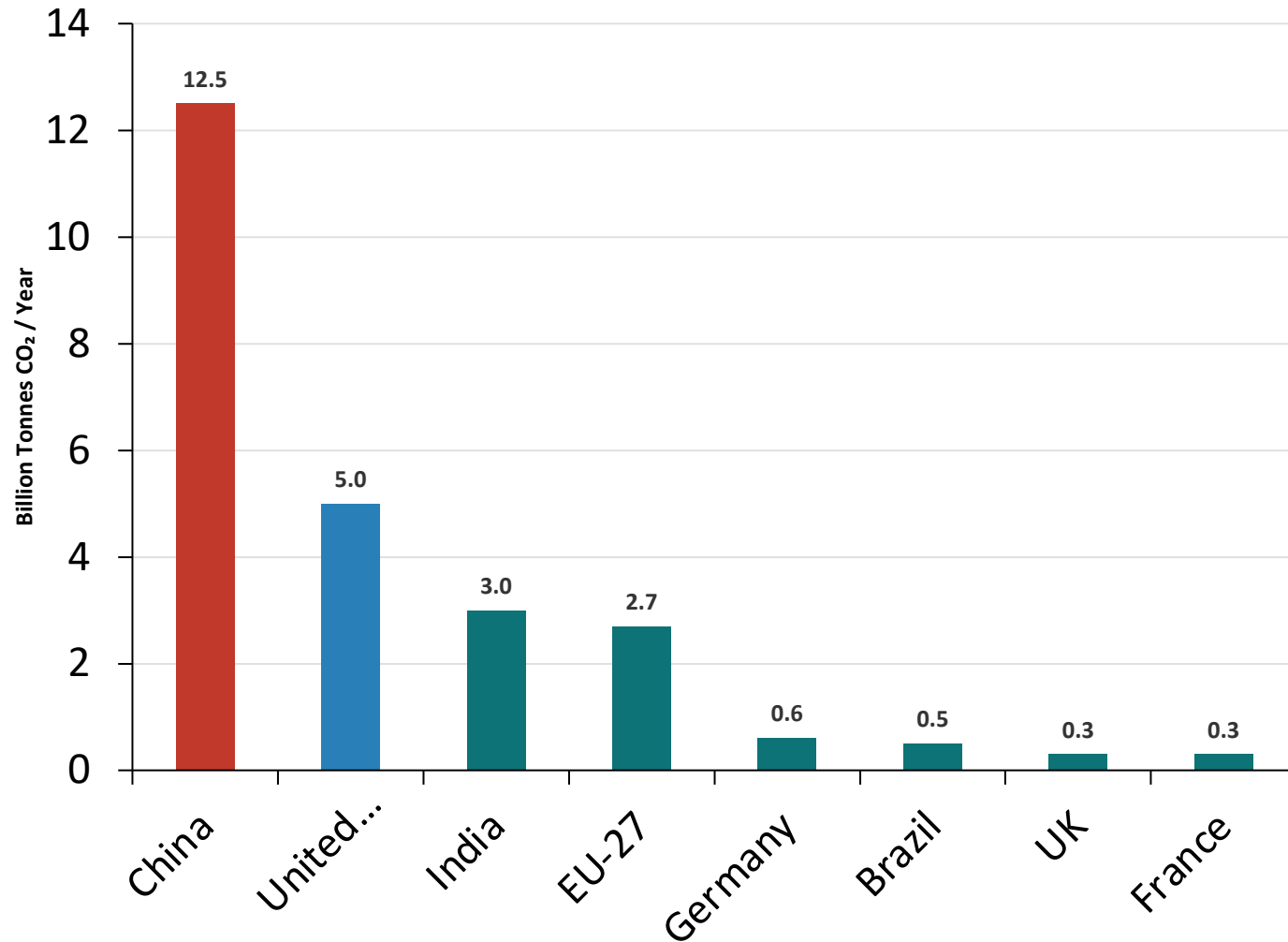
1990: >20 Bt

Emissions nearly 4× from 1950

2024: >35 Bt

Not yet peaked despite slower growth

China Now Emits Over 12 Billion Tonnes — More Than Double the United States



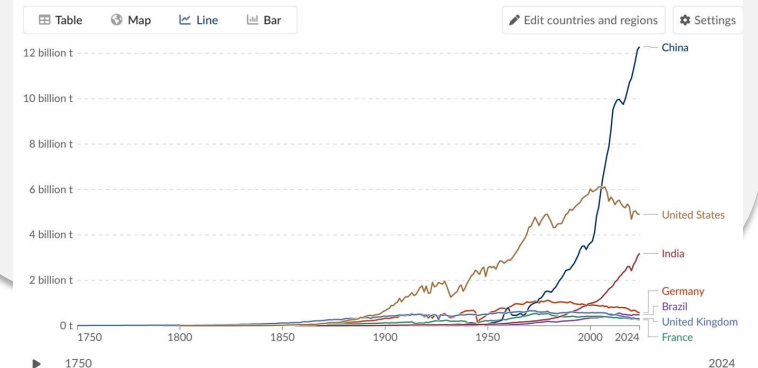
Source: Global Carbon Budget 2025 | figure_3.jpg: Annual CO₂ emissions by country

Key Comparisons

- **China: ~12.5 Bt (~27% of global)**
Slight increase; more than double the US
- **United States: ~5.0 Bt (~14%)**
Declining from peak ~6 Bt (~2005)
- **India: ~3.0 Bt (~8%)**
Rising fast — third-largest emitter
- **EU-27: ~2.7 Bt (~7%)**
Declining; Europe's share has fallen substantially

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

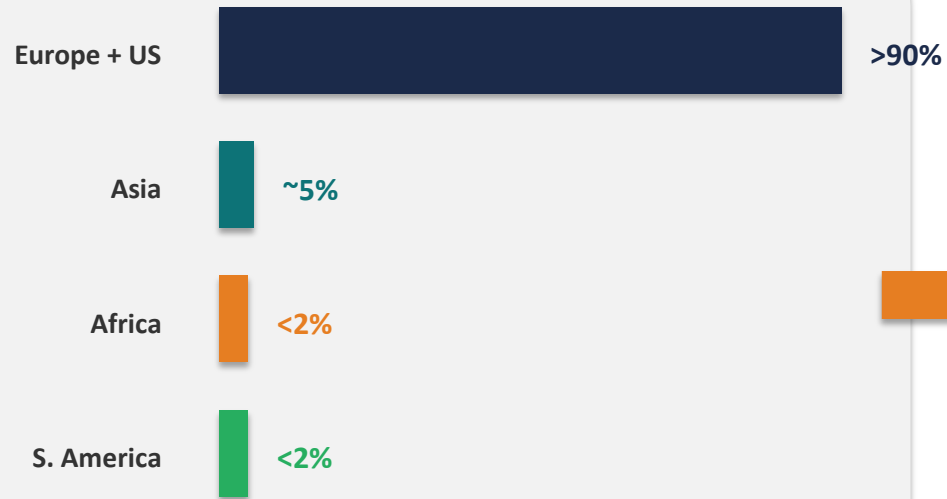


Data source: Global Carbon Budget (2025) — [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Download Share Enter full-screen

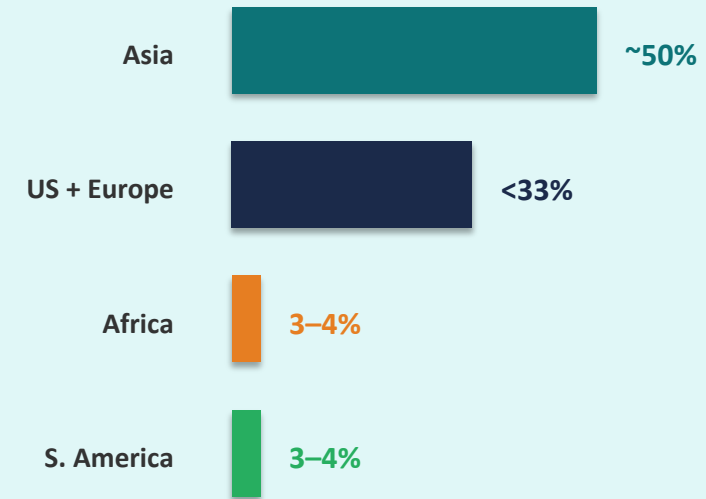
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900: Europe + US Dominate



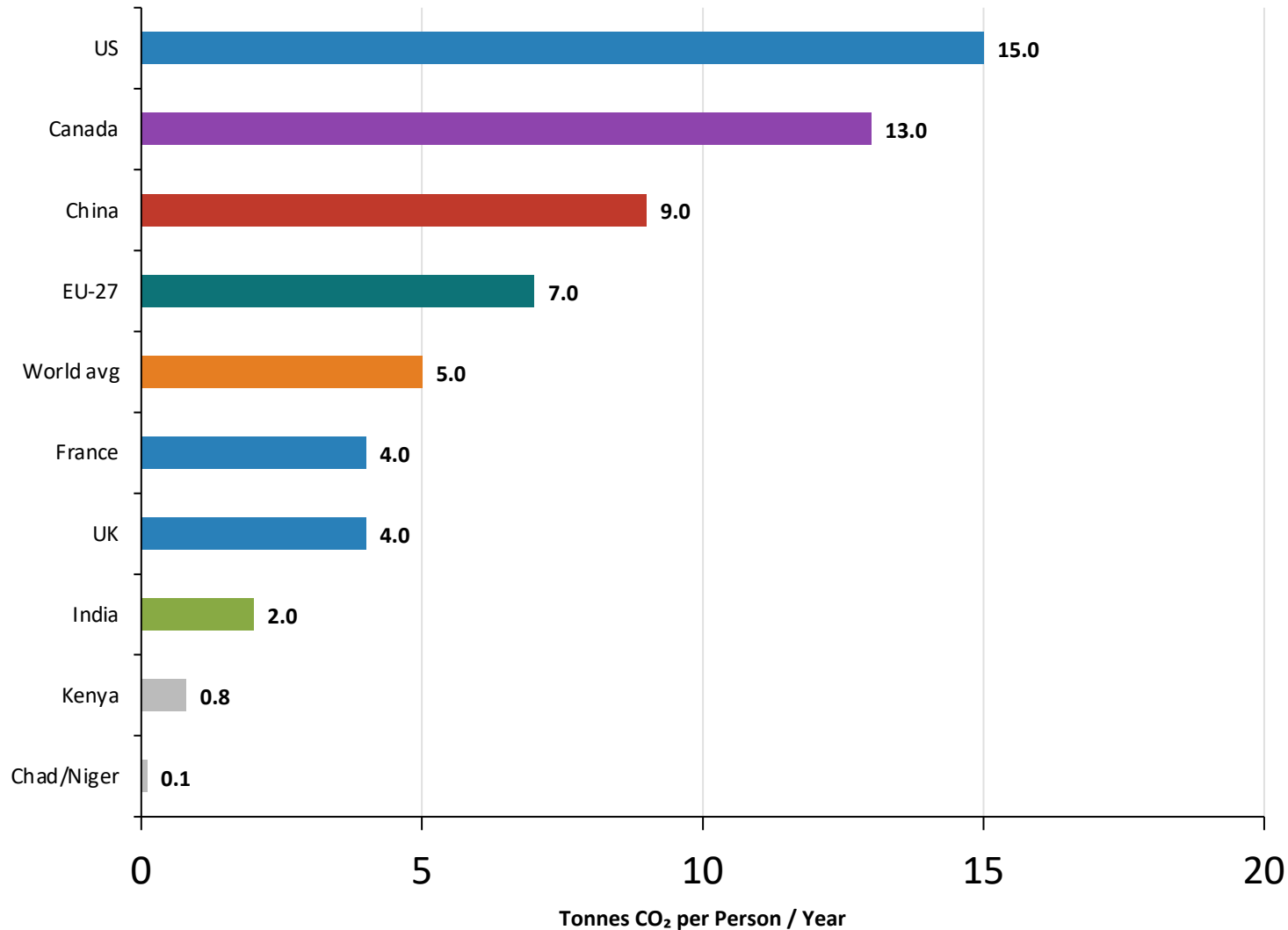
In 1900, Europe and the US accounted for more than 90% of global emissions. By 1950, still over 85%.

2024: Asia Leads the World



Asia — home to nearly 60% of the world's population — now produces roughly half of all emissions. Africa and South America each contribute only 3-4%.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



The 150× Disparity

The average American emits in under two days what the average person in Chad or Niger emits in an entire year.

Among large, populous countries, the US, Australia, and Canada have per capita emissions roughly 3× the global average (~5 t).

The world's highest per capita emitters are major oil-producing nations with small populations (Qatar, UAE, Bahrain, Kuwait).

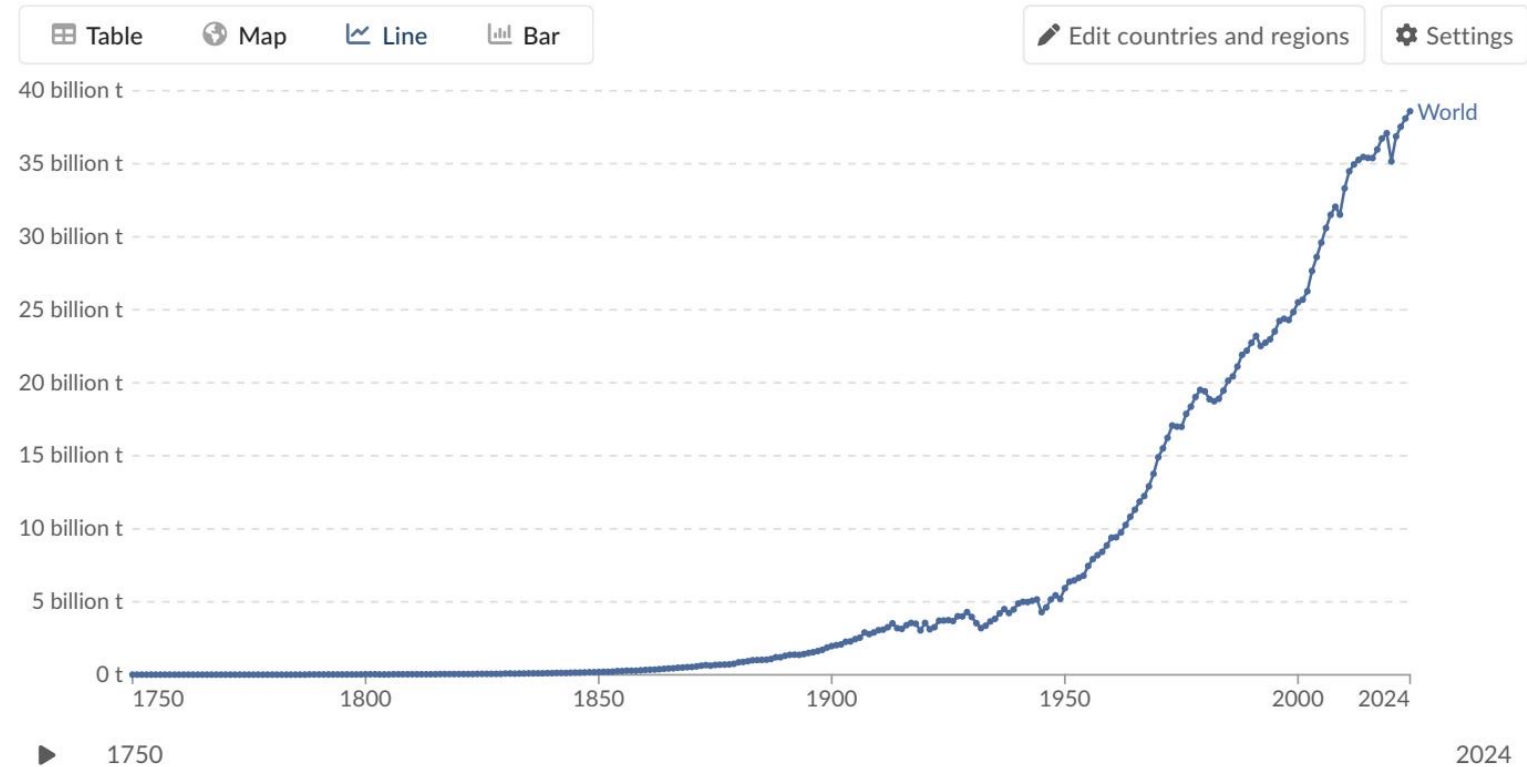
[See figure_1.jpg: Per capita CO₂ emissions over time](#)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

Our World
in Data



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Download Share Enter full-screen

Map Legend

- Decline (–1% to –5%)
- Increase (+2% to +10%)
- Large increase (>+10%)

Key Regional Patterns

- US & much of Europe: declining (–1% to –5%)
- Russia & SE Asia: increasing (+2% to +10%)
- Parts of Africa: mixed, some sharp increases
- Some S. American countries: >+10% increases
- Mixed picture globally: progress in developed economies offset by growth elsewhere

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on energy sources. France and the UK rely more on nuclear and renewables, while Germany and the Netherlands have a higher fossil fuel share.

	France / UK	Germany / Netherlands	Implication
Nuclear Share	High (France ~70%)	Low (Germany phased out)	Nuclear delivers low-carbon baseload
Renewables Share	Growing (UK offshore wind)	Growing but from coal base	Transition pathway matters
Fossil Fuel Share	Lower	Higher (lignite in Germany)	Direct driver of CO ₂ intensity
Per Capita CO ₂	~4 t/person/year	~8–10 t/person/year	2–2.5× difference at similar GDP

KEY INSIGHT

Policy and energy mix choices — not just prosperity — determine per capita outcomes. France’s heavy nuclear investment and the UK’s offshore wind expansion demonstrate that high living standards are compatible with significantly lower per capita emissions.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1

Global emissions exceed 35 Bt and have not yet peaked

Despite slowing growth rates, annual CO₂ from fossil fuels and industry continues to rise.

2

China alone accounts for over 25% of global emissions

~12.5 Bt annually — more than double the United States (~5 Bt). India (~3 Bt) is third and rising fast.

3

Per capita inequalities remain extreme — a 150× gap

The US and Australia at ~15 t/person vs. Chad and Niger at ~0.1 t/person.

4

US and Europe are declining, but historical cumulative responsibility remains large

In 1900, Europe + US >90% of emissions. Their cumulative share remains the largest.

5

Policy and energy mix choices — not just prosperity — determine per capita outcomes

France/UK (~4 t/capita) vs. Germany/Netherlands (~8–10 t/capita): similar GDP, different energy strategies.